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```
clc; clear; close all;
```

Problem #3: U.S. Standard Atmosphere

```
% This problem was solved using AI.  
% AI Tool used was ChatGPT  
  
%AI Summary  
% Temperature, Pressure, and Density Variation with Altitude  
% This script calculates atmospheric properties from 0 to 20,000 m,  
% compares the results with Table C.2, and generates the required plots.
```

Constants

```
g = 9.81;           % Gravitational acceleration (m/s^2)  
R = 287;            % Gas constant for air (J/kg-K)  
  
pa = 101.33;        % Sea-level pressure (kPa)  
Ta = 288.15;        % Sea-level temperature (K)  
  
beta = 0.00650;     % Lapse rate (K/m)  
Tstra = 216.65;     % Stratosphere temperature (K)  
  
pstra = 22.6;       % Pressure at z = 11,000 m (kPa)
```

Altitude Vector

```
z = 0:1000:20000;   % Altitude (m)
```

Initialize Arrays

```
T = zeros(size(z));      % Temperature (K)
p = zeros(size(z));      % Pressure (kPa)
rho = zeros(size(z));    % Density (kg/m^3)
```

Calculate Atmospheric Properties

```
for i = 1:length(z)

    if z(i) <= 11000
        % Troposphere
        T(i) = Ta - beta*z(i);

        p(i) = pa * (1 - beta*z(i)/Ta)^...
            (g/(beta*R));

    else
        % Stratosphere
        T(i) = Tstra;

        p(i) = pstra * exp( ...
            -g*(z(i)-11000)/(R*Tstra) );
    end

    % Density using ideal gas law
    % p must be in kPa and R = 0.287 kJ/kg-K
    rho(i) = p(i)/(0.287*T(i));

end
```

Create Table

```
Temperature_C = T - 273.15;

Results = table(z', Temperature_C', p', rho', ...
    'VariableNames', ...
    {'Altitude_m', 'Temperature_C', 'Pressure_kPa', 'Density_kg_m3'});

disp('Atmospheric Properties from 0 to 20,000 m');
disp(Results);
```

```
Atmospheric Properties from 0 to 20,000 m
```

Altitude_m	Temperature_C	Pressure_kPa	Density_kg_m3
0	15	101.33	1.2253
1000	8.5	89.873	1.1118
2000	2	79.489	1.0066
3000	-4.5	70.098	0.90916
4000	-11	61.627	0.81911
5000	-17.5	54.005	0.73604

6000	-24	47.164	0.65959
7000	-30.5	41.043	0.58936
8000	-37	35.582	0.525
9000	-43.5	30.725	0.46616
10000	-50	26.419	0.41251
11000	-56.5	22.615	0.36372
12000	-56.5	19.301	0.31042
13000	-56.5	16.484	0.26511
14000	-56.5	14.078	0.22642
15000	-56.5	12.024	0.19337
16000	-56.5	10.269	0.16515
17000	-56.5	8.7699	0.14104
18000	-56.5	7.4899	0.12046
19000	-56.5	6.3967	0.10288
20000	-56.5	5.463	0.087861

Table C.2 Data for Comparison

```

z_tab = [0 1000 2000 3000 4000 5000 6000 7000 ...
          8000 9000 10000 15000 20000];

T_tab = [15.00 8.50 2.00 -4.49 -10.98 -17.47 ...
          -23.96 -30.45 -36.94 -43.42 -49.90 ...
          -56.50 -56.50];

p_tab = [101300 89880 79500 70120 61660 54050 ...
          47220 41110 35650 30800 26500 ...
          12110 5529]/1000;           % Convert Pa to kPa

rho_tab = [1.225 1.112 1.007 0.9093 0.8194 ...
            0.7364 0.6601 0.5900 0.5258 ...
            0.4671 0.4135 0.1948 0.08891];

```

Comparison Table

```

fprintf('\nComparison with Table C.2\n');

Comparison = table(z_tab', T_tab', p_tab', rho_tab', ...
                  Results.Temperature_C(ismember(z,z_tab)), ...
                  Results.Pressure_kPa(ismember(z,z_tab)), ...
                  Results.Density_kg_m3(ismember(z,z_tab)), ...
                  'VariableNames', ...
                  {'Altitude_m', ...
                   'T_Table_C2_C', ...
                   'P_Table_C2_kPa', ...
                   'rho_Table_C2', ...
                   'T_Calculated_C', ...
                   'P_Calculated_kPa', ...
                   'rho_Calculated'});

disp(Comparison);

```

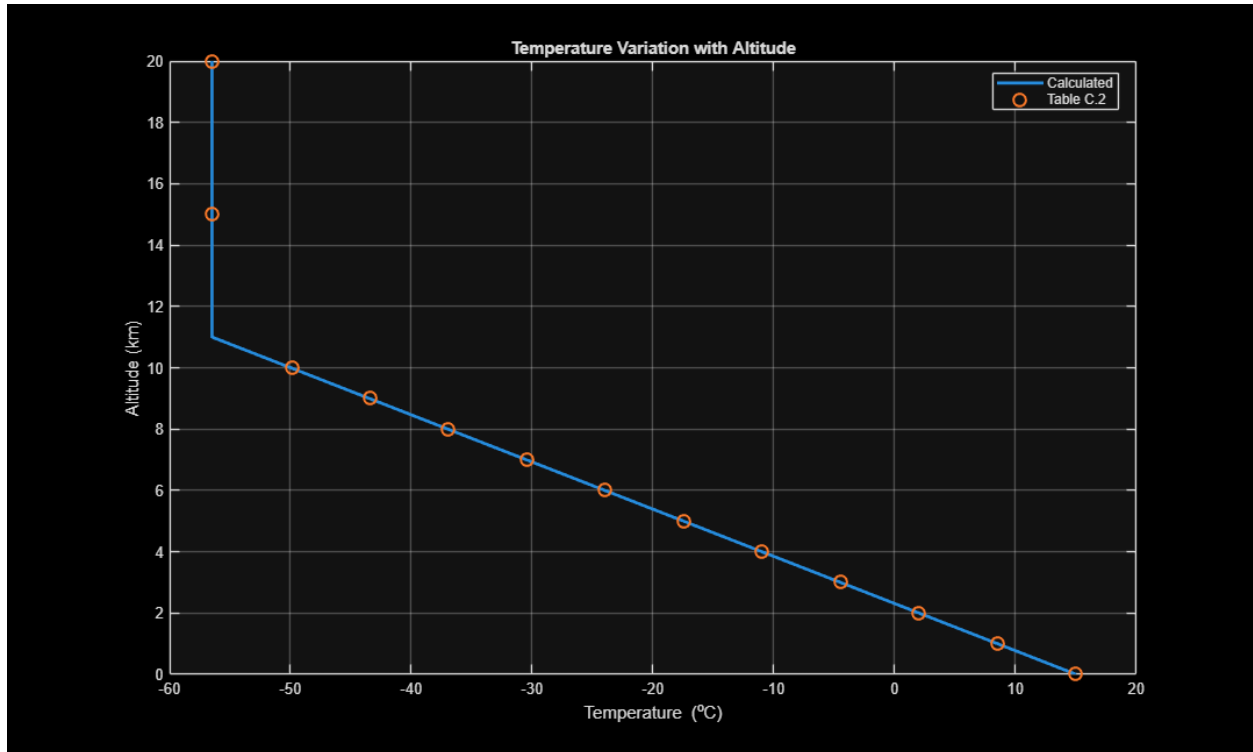
Comparison with Table C.2

<i>Altitude_m</i>	<i>T_Table_C2_C</i>	<i>P_Table_C2_kPa</i>	<i>rho_Table_C2</i>
<i>T_Calculated_C</i>	<i>P_Calculated_kPa</i>	<i>rho_Calculated</i>	
0	15	101.3	1.225
15	101.33	1.2253	
1000	8.5	89.88	1.112
8.5	89.873	1.1118	
2000	2	79.5	1.007
2	79.489	1.0066	
3000	-4.49	70.12	0.9093
-4.5	70.098	0.90916	
4000	-10.98	61.66	0.8194
-11	61.627	0.81911	
5000	-17.47	54.05	0.7364
-17.5	54.005	0.73604	
6000	-23.96	47.22	0.6601
-24	47.164	0.65959	
7000	-30.45	41.11	0.59
-30.5	41.043	0.58936	
8000	-36.94	35.65	0.5258
-37	35.582	0.525	
9000	-43.42	30.8	0.4671
-43.5	30.725	0.46616	
10000	-49.9	26.5	0.4135
-50	26.419	0.41251	
15000	-56.5	12.11	0.1948
-56.5	12.024	0.19337	
20000	-56.5	5.529	0.08891
-56.5	5.463	0.087861	

Plot Temperature vs Altitude

```
figure;
plot(Temperature_C, z/1000, 'LineWidth', 2);
hold on;
plot(T_tab, z_tab/1000, 'o', 'MarkerSize', 8, 'LineWidth', 1.5);

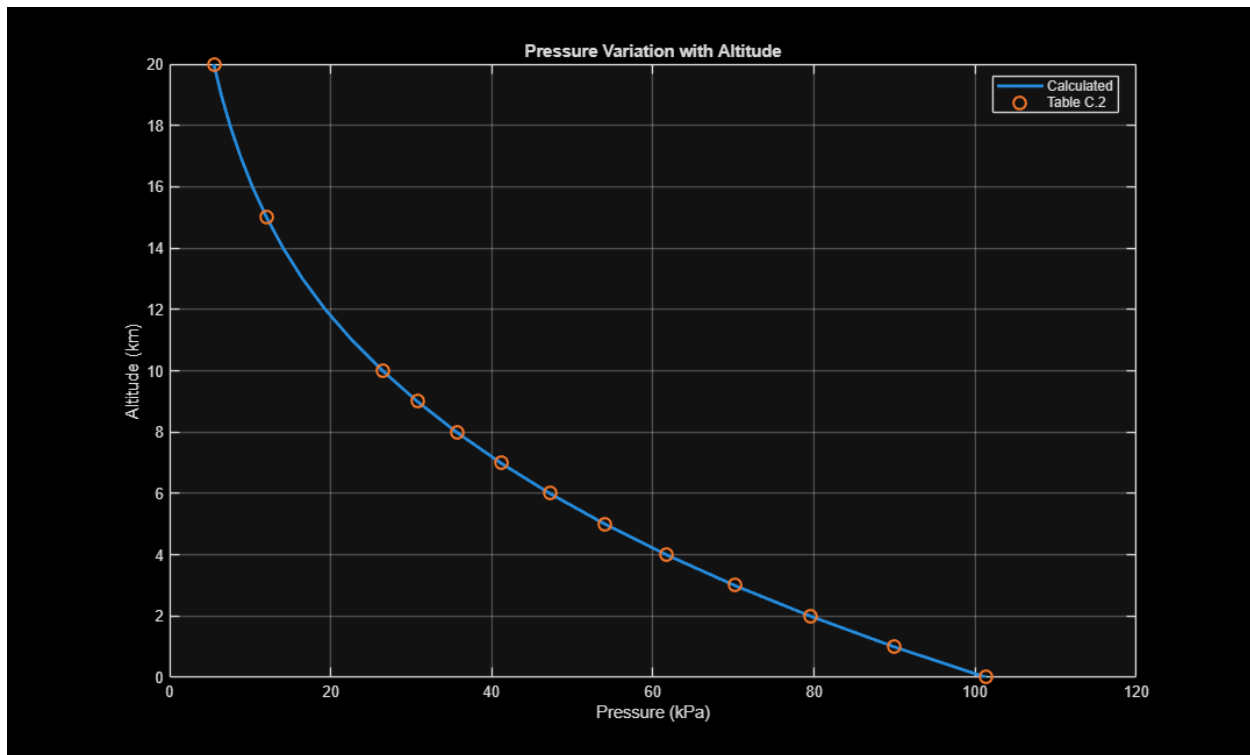
grid on;
xlabel('Temperature (^oC)');
ylabel('Altitude (km)');
title('Temperature Variation with Altitude');
legend('Calculated', 'Table C.2', 'Location', 'best');
```



Plot Pressure vs Altitude

```
figure;
plot(p, z/1000, 'LineWidth', 2);
hold on;
plot(p_tab, z_tab/1000, 'o', 'MarkerSize', 8, 'LineWidth', 1.5);

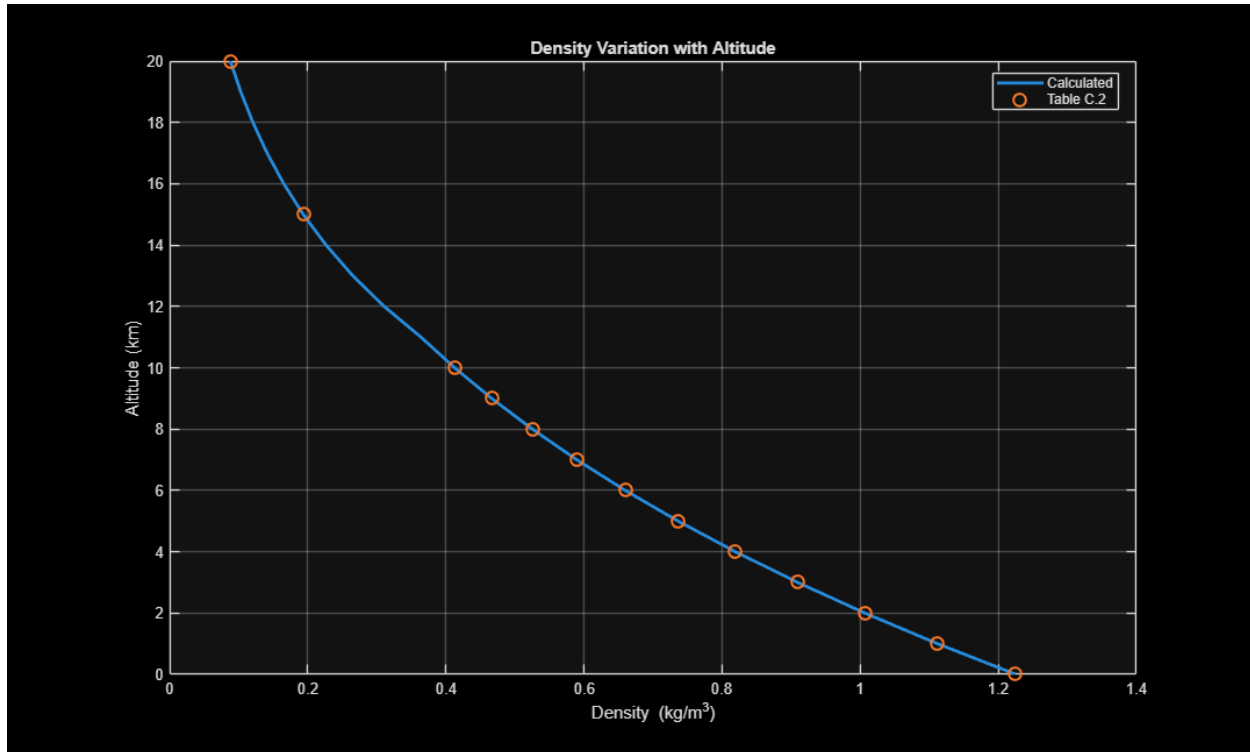
grid on;
xlabel('Pressure (kPa)');
ylabel('Altitude (km)');
title('Pressure Variation with Altitude');
legend('Calculated', 'Table C.2', 'Location', 'best');
```



Plot Density vs Altitude

```
figure;
plot(rho, z/1000, 'LineWidth', 2);
hold on;
plot(rho_tab, z_tab/1000, 'o', 'MarkerSize', 8, 'LineWidth', 1.5);

grid on;
xlabel('Density (kg/m^3)');
ylabel('Altitude (km)');
title('Density Variation with Altitude');
legend('Calculated', 'Table C.2', 'Location', 'best');
```



Percentage Error (Optional)

```
T_calc = Results.Temperature_C(ismember(z,z_tab));
P_calc = Results.Pressure_kPa(ismember(z,z_tab));
rho_calc = Results.Density_kg_m3(ismember(z,z_tab));
```

```
T_error = abs((T_calc - T_tab')./T_tab')*100;
P_error = abs((P_calc - p_tab')./p_tab')*100;
rho_error = abs((rho_calc - rho_tab')./rho_tab')*100;
```

```
ErrorTable = table(z_tab', T_error, P_error, rho_error, ...
    'VariableNames', ...
    {'Altitude_m', 'TempError_percent', ...
    'PressureError_percent', 'DensityError_percent'});
```

```
disp('Percentage Errors Compared with Table C.2');
disp(ErrorTable);
```

```
Percentage Errors Compared with Table C.2
    Altitude_m    TempError_percent    PressureError_percent
DensityError_percent
```

0	0	0.029615
0.023359		
1000	0	0.007421
0.015105		
2000	0	0.01386

0.040046		
3000	0.22272	0.030792
0.015605		
4000	0.18215	0.053283
0.035837		
5000	0.17172	0.083834
0.048346		
6000	0.16694	0.11781
0.077911		
7000	0.1642	0.16239
0.10875		
8000	0.16243	0.19084
0.15197		
9000	0.18425	0.24457
0.20034		
10000	0.2004	0.30614
0.23918		
15000	5.0304e-14	0.71418
0.7337		
20000	5.0304e-14	1.1928
1.1802		

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